THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · MEDIUM LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Sound of two different tuning forks has frequencies 256 Hz and 512 Hz. The wavelength is: 1

(a) equal **(b)** zero

(c) greater for the 512 Hz fork **(d)** greater for the 256 Hz fork

2. A cell has 8 chromosomes. After mitosis each daughter cell has: 1

(a) 4 **(b)** 16 **(c)** 8 **(d)** 2

3. A section of an animal tissue shows flat cells with a single layer. This is likely: 1

(a) cuboidal epithelium **(b)** columnar epithelium

(c) squamous epithelium **(d)** bone

4. A fish differs from an amphibian in having: 1

(a) lungs **(b)** a three-chambered heart

(c) gills throughout life **(d)** moist skin

5. A flower has stamens and a carpel. It is a: 1

(a) bisexual flower **(b)** unisexual flower **(c)** female flower **(d)** male flower

6. A 1 cm rise of the sea level due to melting ice would mostly affect: 1

(a) deserts **(b)** polar caps only

(c) mountain tops **(d)** coastal low-lying areas

7. The density of an object is 2.5 g per cubic cm. In kg per cubic metre this is: 1

(a) 25 **(b)** 250 **(c)** 2500 **(d)** 25000

8. A student has a mixture of sand, salt and camphor. The correct sequence of separation is: 1

(a) evaporation, sublimation, filtration

(b) dissolving, sublimation, evaporation

(c) sublimation, dissolving and filtering, evaporation

(d) filtration, sublimation, evaporation

9. Which of these pairs forms an ionic compound? 1

(a) Mg and Cl **(b)** N and H **(c)** C and H **(d)** C and O

10. Chlorine has isotopes of mass 35 u (75 per cent) and 37 u (25 per cent). The average atomic mass is: 1

(a) 35 u **(b)** 36 u **(c)** 37 u **(d)** 35.5 u

11. A stone thrown vertically up returns to the thrower's hand after 6 s. The velocity of throw is (g = 10): 1

(a) 30 m/s **(b)** 60 m/s **(c)** 15 m/s **(d)** 10 m/s

12. A block of mass 4 kg rests on a rough table. The least force of 12 N makes it just move. Static friction is: 1

(a) 4 N **(b)** 40 N **(c)** 0 **(d)** 12 N

13. In a lever, the effort arm is 40 cm and the load arm is 10 cm. The MA is: 1

(a) 0.25 **(b)** 50 **(c)** 4 **(d)** 30

14. Two musical instruments produce notes of the same pitch and loudness but you can tell them apart. This is due to differing: 1

(a) amplitude **(b)** speed
(c) quality or timbre **(d)** frequency

15. The organelle which is found in all eukaryotic cells but whose number is highest in actively respiring cells: 1

(a) cell wall **(b)** mitochondria **(c)** nucleus **(d)** vacuole

16. A person can bend the arm at the elbow only in one plane. This joint is a: 1

(a) hinge joint **(b)** fixed **(c)** ball and socket **(d)** pivot joint

17. The correct sequence of taxonomic ranks for the tiger from the highest is: 1

(a) Chordata, Animalia, Mammalia, Carnivora
(b) Animalia, Mammalia, Chordata, Felidae, Carnivora, Panthera
(c) Animalia, Chordata, Mammalia, Carnivora, Felidae, Panthera, tigris
(d) Panthera, Felidae, Carnivora, Mammalia

18. The number of chromosomes in a human zygote is: 1

(a) 46 **(b)** 44 **(c)** 23 **(d)** 92

19. In the water cycle, the Sun provides energy for: 1

(a) evaporation **(b)** run-off **(c)** precipitation **(d)** condensation

20. A runner covers 100 m in 12.5 s. The estimated speed is: 1

(a) 80 m/s **(b)** 12.5 m/s **(c)** 8 m/s **(d)** 1.25 m/s

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. A force of 40 N moves a body through 5 m in the direction of the force. Find the work done and the KE gained if the body starts from rest with mass 4 kg. 2

22. The frequency of a wave is 20 Hz and its wavelength 17 m. Find the speed and the time for 10 oscillations. 2

23. Why do red blood cells burst when placed in distilled water but not plant cells? 2

24. Why are the cells of the meristematic tissue closely packed with no intercellular spaces? 2

25. A student has a sample of a unicellular organism with a nucleus and cilia. In which kingdom will you place it and why? 2

26. Why do plants that are propagated vegetatively not show variation? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. A potato cube weighing 10 g is placed in distilled water and another in strong salt solution. Predict the masses after some time and explain. 3

28. Describe how the guard cells control the opening and closing of stomata. 3

29. Classify the following into the correct groups: banyan, fern, Spirogyra, moss, pine, rice. 3

30. Differentiate between the structure and functions of a sperm and an ovum. 3

31. Explain the effect of albedo on the climate with the three surfaces: snow, forest, ocean, using values 0.85, 0.1, 0.06. 3

OR

Explain the importance of scientific notation, using examples of the size of an atom and the distance to the Sun. 3

32. A mixture of NaCl and KNO3 is dissolved in hot water and cooled. Which crystallises first and why? 3

OR

State three differences between sodium chloride and carbon tetrachloride. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. A student cuts a thin section of a young stem and observes it under a microscope. Describe the steps and name the tissues seen from outside to inside, with the function of each. 5

34. A student collects ten organisms: mushroom, rose, whale, bacterium, Euglena, fern, snake, pigeon, earthworm and moss. Classify each into the kingdom and the group. 5

35. Describe the sequence of events from the production of gametes to the birth of a baby in humans, with the organs involved. 5

36. The Sun's radiation of 1000 W/m squared falls on a surface of 2 square metres. Find the energy received in 1 hour and the energy absorbed by
(a) snow with albedo 0.85 and
(b) forest with albedo 0.1 in kWh. 5

OR

Explain how scientists establish a theory (e.g. cell theory or atomic theory) through evidence over time with at least five steps. 5

37. Explain, with reasons, how air is separated in industry, and the differences in the boiling points used. 5

OR

A student heats 6.0 g of copper in air. The mass of the product copper oxide is 7.5 g. Find the mass of oxygen, the ratio of Cu : O and compare with the formula CuO (Cu = 63.5, O = 16). 5

38. Describe the isotopes of carbon, their composition, use and the idea of carbon dating. 5

OR

A driver at 72 km/h sees an obstacle 50 m ahead and brakes with a retardation of 5 m/s squared after a reaction time of 0.5 s. Will he stop in time? 5

END OF QUESTION PAPER — Total: 80 Marks